ADVANCED ENERGY MATERIALS
www.advenergymat.de

Advancing Humidity-Resistant Triboelectric Nanogenerators Through MoS_2-Encapsulated SiO_2 Nanoparticles for Self-Powered Gas Sensing Applications

Do-Heon Kim, Ji Young Park, Han Sol Choi, Jeonghoon Cho, Hyun Soo Kim, Jeong Eun Mo, Jin-Kyeom Kim, Tae Kyoung Yoon, Sung Hun Hur, Jae Joon Kim, Hye Sung Park, Hyun Cheol Song,* and Jeong Min Baik**

In this study, the humidity-resistant triboelectric nanogenerators (TENGs) utilizing MoS_2-encapsulated SiO_2 nanoparticles (NPs), aimed at enhancing self-powered gas sensing applications, are reported. The core-shell structure, featuring a thin MoS_2 layer uniformly grown on SiO_2, addresses common humidity-induced performance degradation. The growth mechanism involves the decomposition and sulfidation of molybdenum species, with MoS_2 selectively nucleating on SiO_2 to form a stable, hydrophobic shell. This MoS_2 layer effectively shields the SiO_2 interface from water molecule penetration, thus stabilizing charge density and significantly reducing charge decay, even under high humidity conditions. TENGs constructed with these core-shell NPs exhibit high triboelectric charge density and exceptional durability, retaining more than 70% output over 25 h at 99% relative humidity (RH). Furthermore, the fabricated TENG reliably powers a gas sensor array, enabling accurate gas detection in extreme humidity. This work demonstrates the potential of MoS_2-encapsulated SiO_2 TENGs as robust, self-powered energy solutions for environmental monitoring and wearable devices in challenging humidity conditions.

1. Introduction

Energy harvesters are widely studied as environmentally friendly and sustainable energy sources due to their flexibility and adaptability in terms of output, size, and application environments.[1,2] Micro-energy harvesters (MEH), such as thermoelectric,[3] magnetostrictive,[4] and piezoelectric generators,[5,6] have emerged as efficient energy solutions for Internet of Things (IoT) systems and wearable devices. These technologies address the growing challenges posed by global warming, such as increased wildfires and natural disasters, as well as evolving healthcare needs in aging societies.[7–10]

Among these, triboelectric nanogenerators (TENGs) stand out for their exceptional capability to convert diverse forms of mechanical energy into electrical energy, operating with minimal environmental limitations and eliminating the need for additional converters.[11–13] TENGs generate electricity through contact electrification (CE), a process in which two materials with distinct charge affinities come into contact. During this process, electrons or ions transfer across the surfaces to equalize the electrostatic potential, creating a potential difference between two electrodes.[14–16] Over the past decade, research advancements

D.-H. Kim, J. Y. Park, H. S. Choi, J. E. Mo, T. K. Yoon, H. C. Song, J. M. Baik
School of Advanced Materials Science and Engineering
Sungkyunkwan University (SKKU)
Suwon 16419, Republic of Korea
E-mail: hcsong@kist.re.kr; jbaik97@skku.edu

J. Cho, J. J. Kim
Department of Electrical Engineering
Ulsan National Institute of Science and Technology
Ulsan 44919, South Korea

H. S. Kim, S. H. Hur, H. C. Song
Electronic Material Research Center
Korea Institute of Science and Technology (KIST)
Seoul 02792, Republic of Korea

J.-K. Kim
Department of Materials Science and Engineering
University of Wisconsin-Madison
Madison, WI 53706, USA

J.-K. Kim
Research Center for Advanced Materials Technology (RCAMT)
Sungkyunkwan University
2066 Seobu-ro, Jangan-gu, Suwon, Gyeonggi-do 16419, Republic of Korea

H. S. Park
KU-KIST Graduate School of Converging Science and Technology
Korea University
145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea
E-mail: hyesungpark@korea.ac.kr

H. C. Song, J. M. Baik
KIST-SKKU Carbon-Neutral Research Center
Sungkyunkwan University (SKKU)
Suwon 16419, Republic of Korea

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/aenm.202405278

DOI: 10.1002/aenm.202405278

 © 2025 Wiley-VCH GmbH

have introduced novel triboelectric materials with high strength, chemical resistance, and enhanced charge density.[17–20] In addition, innovations in structural design,[21] circuit management,[22,23] and mechanical systems[24,25] have broadened the output potential of TENGs, extending their utility not only as energy sources, but also across diverse application fields. The lightweight and miniaturized nature of TENGs positions them as promising power solutions for mobile healthcare devices and the IoT.[26–28] Flexible TENGs have facilitated applications in bio-implantable devices.[29] One notable application leverages high-voltage characteristics of TENGs, enabling their use in plasma generation,[30,31] photonic modulators,[32] and beyond.

However, rapid charge dissipation remains a challenge that must be addressed to achieve widespread commercialization of TENGs. The transferred electrons on the surface of tribo-materials naturally dissipate due to air breakdown, as explained by Paschen's law.[33] In conditions of high relative humidity (RH), tribo-charges dissipate more quickly, following a percolation model, as the adsorbed water layer on the surface increases electrical conductivity.[34,35] While the water-bridge on the contact surface aids CE through ion transfer, ions cannot remain trapped on the surface if the water layer exceeds 2 nm in thickness.[36,37] Furthermore, the adsorbed water layer acts as a barrier to both the effective electric field generated by the tribo-charge and the charge transfer process itself.[38,39] Adverse effects such as corrosion and swelling of the tribo-material, accelerated by absorbed water, alongside reduced mechanical strength and an unstable contact area, also significantly decrease TENG output performance.[40] Consequently, under high humidity conditions up to RH 99%, the charge density of conventional tribo-materials decreases to less than 5%, with a notable increase in the charge decay rate due to absorbed moisture.[41–43]

To address performance degradation caused by high humidity, significant advancements in TENG technology have been driven by extensive innovations. For instance, the DOS matching strategy achieved a high current density of 1333 mA m^{-2} even under 100% RH by introducing conductive contact layers, which facilitated electron transfer within the tribo-layer while minimizing charge leakage into the air.[44] Similarly, the charge excitation strategy demonstrated humidity resistance up to RH 90% and durability over 1 000 000 cycles by harvesting current through a variable capacitor (C_v) charged by an external TENG, where C_v being minimally affected by humid conditions.[45] However, the development of tribo-materials that can inherently maintain charge stability under environmental disturbances remains a critical challenge. This is not only vital for improving TENGs, but also for offering robust solutions in diverse modern triboelectric applications, including air filters,[46] textile sensors,[47] and wound healing dressings.[48] Consequently, significant research has been dedicated to developing humidity-resistant materials, including the creation of hydrophobic surfaces,[49,50] fibrous architectures for moisture dissipation,[51,52] and multilayered structures.[53] Hydrophobic surface modification has most widely been explored to prevent penetration of water molecules, through physical roughness modifications[54] and chemical surface functionalizations such as fluorination[55,56] and amino treatment.[50] However, surface charge densities of hydrophobic tribo-materials remain relatively low due to inherent limitations of the triboelectric effect.[39] Fibrous tribo-materials can readily dissipate moisture through evaporation, exhibiting droplet- and sweat-resistance; however, their output maintenance through continuous evaporation is extremely limited in highly humid environments.[57] Multilayered TENGs capable of retaining over 70% of their output under high humidity have recently been developed by engineering the interface band structure or incorporating engrafted trapped sites.[58,59] Nevertheless, even minimal moisture penetration within the multilayer structure can lead to rapid dissipation of local space charges, compromising long-term stability. Recently, water-reactive tribo-materials have been developed that leverage humid conditions by transferring surface mobile ions via water bridges or activating hydrogen bonding, which enhances tribo-positivity.[60–62] However, the charge density of these materials is fundamentally limited to several tens of μC/m^2 due to the high hygroscopicity associated with hydrogen bonding, which leads to significant charge dissipation, and the restriction of ionic polarization imposed by the architectural limits of doping concentration.[63,64] Therefore, further improvements are necessary to enhance the charge density and maintain it over extended periods in harsh environments. TENGs with stable high outputs can be designed to operate reliably in any environment and will be quickly adapted for various applications.[65,66]

Here, we address humidity-related challenges by fabricating a thin MoS_2 layer that uniformly encapsulates SiO_2. Through selective nucleation, a stable, continuous MoS_2 layer forms on the SiO_2 surface, where molybdenum species are reduced and sulfided. This shell effectively isolates the SiO_2 interface from water molecules, enhancing charge retention and significantly reducing charge decay, even at 99% RH. Furthermore, the MoS_2 encapsulation generates an electric field at the core-shell interface, facilitating the migration of positive charges from the MoS_2 surface to the SiO_2 surface and enhancing charge transfer. This results in increased charge density, contributing to the superior electrical output of TENG and durability. The resulting TENG, powered by SiO_2@MoS_2 nanoparticles (NPs), demonstrated high stability and performance, effectively powering a self-sustaining gas sensing system capable of distinguishing four gases (H_2, SO_2, CO, and H_2S) at RH 99%.

2. Result and Discussion

Our primary objective is to develop a novel tribo-material with high humidity resistance, achieved solely through nanostructuring, without the need for multilayer stacking or additional packaging. Our strategy focuses on engineering a core-shell composite structure, where a thin MoS_2 layer uniformly encapsulates SiO_2. As shown in **Figure 1a**, this structure aims to simultaneously benefit from the low decay rate of charges trapped at the interface and the inherent isolation of charges from water molecules.[67–69] To completely inhibit the penetration of water molecules into the MoS_2-SiO_2 interface and ensure a large specific interface area, MoS_2 was synthesized via a one-step hydrothermal method, allowing for its direct growth onto

J. M. Baik
SKKU Institute of Energy Science and Technology (SIEST)
Sungkyunkwan University
Suwon 16419, Republic of Korea

 © 2025 Wiley-VCH GmbH

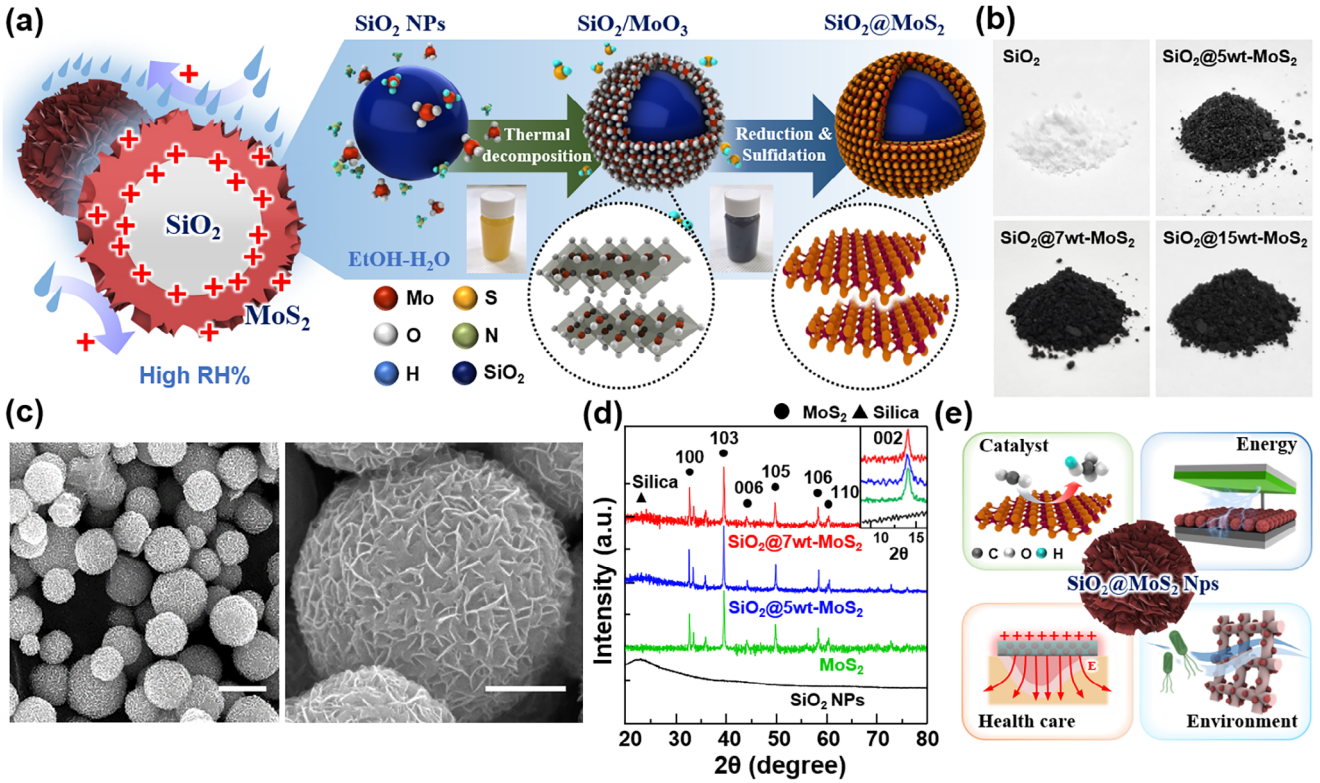


Figure 1. a) Schematic representation of humidity-resistive SiO_2@MoS_2 core-shell NPs and hydrothermal synthesis process. b) Photographs of as-prepared NPs with controlled MoS_2 weight ratio. c) SEM images of the 7 wt.% SiO_2@MoS_2 NPs. Scale bar: 500 nm left side image and 100 nm right side image. d) XRD patterns of as-prepared NPs with controlled MoS_2 weight ratio. e) Potential applications of SiO_2@MoS_2 NPs through integrated material system.

the SiO_2 surface. The mechanism underlying the formation of SiO_2@MoS_2 core-shell NPs is shown in Figure 1a. Detailed parameters and synthesis process are described in the Experimental Section. Initially, the conventional Stöber method was employed to synthesize SiO_2 NPs with a diameter of ≈150–200 nm. Following this, an MoS_2 shell was formed on the silica surface in a 1:1 EtOH-H_2O solution at a mild temperature of 180 °C. To the best of our knowledge, the possible reaction pathway for the synthesis of MoS_2 shell is proposed as follows:

$$(NH_4)_6Mo_7O_{24}.4H_2O \rightarrow 6NH_3+7MoO_3+3H_2O \quad (1)$$

$$HSCH_2CHNH_2COOH+H_2O \rightarrow CH_3COCOOH + NH_3 + H_2S \quad (2)$$

$$MoO_3+3H_2S+H_2O \rightarrow MoO_2+SO_4^{2-} + H^+ \quad (3)$$

$$MoO_2+2H_2S \rightarrow MoS_2+SO_4^{2-} + H^+ \quad (4)$$

The reaction process involves the formation of MoO_3 through the thermal decomposition of ammonium molybdate tetrahydrate (1) and the release of H_2S from L-cysteine (2). The entire process includes the subsequent reduction (3) and sulfidation (4) of MoO_3 to MoS_2 in the presence of H_2S. Under precisely controlled temperature and pressure conditions, heterogeneous nucleation of MoS_2 predominantly occurs at the numerous nucleation sites provided by SiO_2. As shown in Figure 1a, during the first few hours of the hydrothermal process, a yellow-green solution is still visible, which gradually transitions to a deep black color, indicating the formation of MoS_2. After calcination to remove residual organic materials, the resulting core-shell particles exhibit a complete matte black appearance when more than 7 wt.% of MoS_2 precursor is added to SiO_2 (denoted as SiO_2@7wt-MoS_2), as shown in Figure 1b. The detailed morphology of the synthesized SiO_2@MoS_2 core-shell structure was analyzed using field-emission scanning electron microscopy (FE-SEM). As shown in Figure 1c, the MoS_2 shell was successfully encapsulated on all silica NPs, with no exposed silica surface. The MoS_2 exhibited a distinct flower-like surface morphology, indicative of a large specific surface area. Even after the shell formation, the NPs retained their overall spherical shape, confirming that a uniform MoS_2 shell had formed on the silica surface. Typically, higher supersaturation of the solvent results in the formation of a greater number of smaller-sized crystal particles.[70,71] The addition of ethanol reduces the dielectric constant of the precursor solution, lowering solubility and consequently increasing supersaturation.[72] This leads to smaller MoS_2 crystallite sizes on the surface of the core-shell NPs, which, in turn, facilitates uniform shell growth. Moreover, silica NPs dispersed in the 1:1 EtOH-water solvent exhibit a thick solvation film and enhanced dispersion stability.[73] As a result, each SiO_2 particle remains separate prior to the formation of the MoS_2 shell, promoting the distinct formation of individual SiO_2@MoS_2 core-shell NPs without aggregation, as shown in Figure 1c and Figure S1a (Supporting Information). In the absence of ethanol, hydrothermal synthesis

© 2025 Wiley-VCH GmbH

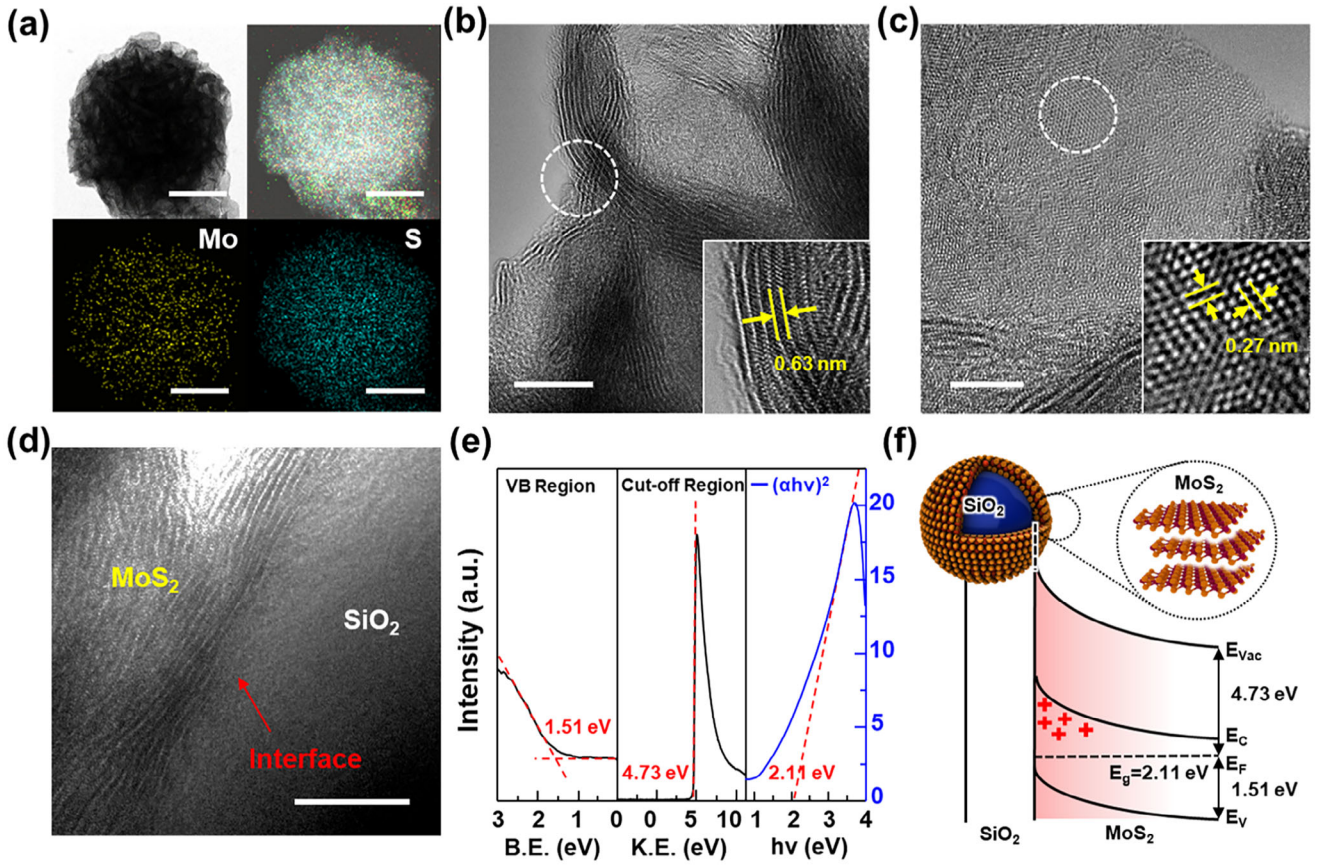


Figure 2. a) STEM-EDS mapping images and elemental distribution of SiO_2@MoS_2: region of mapping, Mo and S. Scale bar: 100 nm mapping images. b) Cross sectional HRTEM image of MoS_2 shell, lattice spacing of 0.63 nm which correspond to (002) plane of MoS_2. Scale bar: 10 nm. c) HRTEM image of MoS_2 shell, lattice pattern in agreement with 2H-ABAB stacked MoS_2. Scale bar: 5 nm. d) Cross sectional HRTEM image of SiO_2@MoS_2 interface. Scale bar: 10 nm. e) UPS VB region, UPS secondary electron region and Tauc plot using formula for direct band gap of as-prepared MoS_2 shell. f) Schematic illustration of band diagram across the SiO_2@MoS_2 core-shell NPs.

under the same conditions resulted in the agglomeration of particles and non-uniform shells, as observed in Figure S1b (Supporting Information). Additionally, at high temperatures above 180 °C, homogeneous nucleation of MoO_3 predominantly occurred resulting in the formation of separate MoS_2 NPs (Figure S1c,d, Supporting Information), and at lower temperature below 180 °C, the MoS_2 shell did not formed (Figure S1e,f, Supporting Information). The crystal structure of the as-prepared NPs was characterized by X-ray diffraction (XRD). Figure 1d shows the XRD patterns of the synthesized SiO_2 NPs, MoS_2 NPs, and SiO_2@MoS_2 NPs with controlled wt% of MoS_2 precursors. The MoS_2 NPs were synthesized under the same hydrothermal conditions, excluding the presence of silica. The characteristic peaks of MoS_2 were observed at 14.40°, 32.89°, 39.28°, 44.07°, 49.57°, 58.47°, and 60.79°, corresponding to the (002), (100), (103), (006), (105), (106), and (110) planes. These sharp peaks were measured for both MoS_2 NPs and all SiO_2@MoS_2 NP samples, confirming the MoS_2 hexagonal crystal structure (2H-MoS_2, JCPDS-37-1492) without impurities.[74,75] A broad peak related to silica was observed at $\approx 2\theta$–23° in the pattern of the SiO_2@MoS_2 core-shells, unlike in the bare MoS_2 NPs. Furthermore, the presence of sharp (100) reflections indicates in-plane crystallinity.[76] The SiO_2@MoS_2 core-shell is fully synthesized as intact particles and subsequently applied to the system, enabling the extension of their integration into various forms such as composites, fillers, decoration, and films. This compatibility enables applications in diverse fields such as wound healing patches, contamination filters, catalyst, and energy harvesters, which are based on triboelectricity and electret materials.

The detailed structure of the SiO_2@MoS_2 NPs was characterized using high-resolution TEM (HR-TEM). **Figures** 2a and S2a (Supporting Information) illustrate the spherical overall morphology of the SiO_2@MoS_2 core-shell NPs. The corresponding energy dispersive spectroscopy (EDS) spectrum primarily shows the presence of Mo, S, Si, and O elements. The calculated atomic ratio of Mo and S (1:1.99) closely matches the stoichiometric composition of MoS_2, as depicted in Figure S2b (Supporting Information). This suggests that, under our experimental conditions, intermediate phases such as MoO_3 and MoO_2 are scarcely present. Figure 2b,c provides the enlarged HR-TEM images that show the detailed information about the contact surface and crystal structure of the NPs. In Figure 2b, the MoS_2 exhibits a 2D structure, and the fine, closely packed lines visible in the image represent the individual layers of MoS_2. These layers are stacked with

© 2025 Wiley-VCH GmbH

an interlayer spacing of ≈0.63 nm, corresponding to the (002) planes of MoS_2. The clear observation of these layers indicates high crystallinity, suggesting the formation of MoS_2 nanosheets or nanoflowers. The lattice spacing of the MoS_2 shell, measured from the enlarged HR-TEM image in Figure 2c, shows a distance of 0.27 nm for the MoS_2 (100) plane, with an angle of 60° between them. These lattice patterns are consistent with the semiconducting 2H-ABAB layer-stacked MoS_2, which aligns well with the XRD patterns shown in Figure 1d and the selected area electron diffraction (SAED) patterns from the HR-TEM images in Figure S2a (Supporting Information).

To further investigate the growth mechanism of MoS_2 on SiO_2, the MoS_2-SiO_2 interface was closely examined. As shown in Figure 2d, the distinct interface reveals a well-defined boundary between the MoS_2 shell and the SiO_2 core, indicating the successful formation of the core-shell structure. The parallel layers of MoS_2 extending from the interface are clearly visible, highlighting the uniform growth of the MoS_2 shell on the SiO_2 surface. To the best of our knowledge, this suggests that precisely controlled hydrothermal condition along with the elevated supersaturation by ethanol promotes the uniform development of MoS_2 crystallite onto the surface of the SiO_2 core. The MoS_2 formed on the surface continuously grew, expanding and adhering during the hydrothermal reaction, consequently forming randomly oriented MoS_2 nanoflowers on the outer surface.[77,78] Figure 2d confirms the strong adhesion of the MoS_2 to the SiO_2 core, achieving the desired core-shell architecture with a clear and well-defined interface. In addition, the MoS_2 shell thickness of the core-shell NPs was measured to be ≈43 nm by subtracting the average size of the SiO_2 NPs from that of the SiO_2@MoS_2 core-shell NPs, as shown in Figure S3 (Supporting Information). The small standard deviation of 2.49 in the MoS_2 shell thickness, derived from the particle size distributions of SiO_2 and SiO_2@MoS_2, indicates the uniformity of the shell thickness.

The band alignment and energy levels at the interface between MoS_2 and SiO_2 are critical for understanding charge transfer behavior and charge trap capability. Figure 2e,f explores the electronic band structure of the SiO_2@MoS_2 core-shell system, integrating the ultraviolet photoelectron spectroscopy (UPS) data along with UV–vis absorption spectrum results. The extraction of the band structure using UPS plots for the secondary electron cut-off region and valence band (VB) region was conducted with an applied bias of 10 V. The calculated work function, determined by subtracting the cut-off energy from the excitation energy ($h\nu = 21.22$ eV, He I source) and applied bias, was 4.73 eV. Additionally, the energy difference between the Fermi level (E_F) and the VB maximum (E_V) was found to be 1.51 eV. The linear extrapolation of the $(\alpha h\nu)^2$–hν plot (Tauc plot) indicates an optical bandgap of 2.11 eV, confirming the n-type semiconducting nature of MoS_2. Based on these results, the band alignment between MoS_2 and SiO_2 was estimated, showing upward band bending near the MoS_2-SiO_2 interface, as depicted in Figure 2f. To the best of our knowledge, this upward band bending might be attributed to the influence of intrinsic sulfur vacancy-related defects and the formation of active sites due to the incomplete elimination of absorbed gas molecules.[79–81] In the SiO_2-MoS_2 heterojunction, where the band bending region within MoS_2 extends over at least 100 nm, the MoS_2 shell of core-shell NPs with an average thickness of 43 nm is entirely within this regime.[59] Consequently, the positive charge is transported to the interface by the built-in electric field and subsequently stored on the surface of SiO_2.

The SiO_2@MoS_2 NPs were spin-coated onto a Ni-mesh electrode supported by a 2 × 2 cm^2 acrylic plate (**Figure 3a**). Ni-mesh was selected as the electrode to achieve a high specific surface area as well as the characteristic stretchability and mechanical stability of fiber-based metal meshes.[82,83] After heat treatment at 60 °C for 3 h, the NPs formed a durable film on the Ni-mesh as a positively charged framework (denoted as Ni/SiO_2@MoS_2), as illustrated in Figure 3a. Perfluoroalkoxy alkane (PFA), a highly effective anionic material, was attached onto the top Al electrode, which functioned as the negatively charged layer. The fabricated TENG was placed in a pushing tester under ambient conditions of RH 24% and a temperature of ≈22 °C. The open-circuit voltage (V_{OC}) and short-circuit current (J_{SC}) of the TENGs were measured under a cyclic compressive force of ≈30 N at a frequency of 3 Hz, as shown in Figure 3b and Figure S4 (Supporting Information). The output exhibited an optimal point due to opposing tendencies: as the MoS_2-to-SiO_2 ratio decreases, the specific MoS_2-SiO_2 interface area increases while the built-in electric field decreases due to shallow upward band bending. Therefore, the TENG based on the SiO_2@MoS_2 NPs with 7 wt.% MoS_2 exhibited optimal AC-type instantaneous electric output performance of 750.3 V and 45.5 μA, compared to 450.5 V and 15.1 μA for the reference TENG based on Ni-mesh. This improvement can be attributed to the increase in charge density generated via CE. The TENG based on Ni-mesh showed a charge density of only 83 μC m^{-2}, whereas the SiO_2@MoS_2-coated Ni-mesh exhibited a significantly higher charge density of 471 μC m^{-2}. Based on the band diagram shown in Figure 2f, this enhancement can be explained by the improved charge transfer, driven by the lower work function of MoS_2 (4.73 eV), and the migration of positive charges to the MoS_2–SiO_2 interface due to the generation of a built-in electric field. Furthermore, SiO_2 is a well-known electret material, extensively used in microelectromechanical systems (MEMS) due to its high charge capacity.[84,85]

This outstanding performance of the SiO_2@MoS_2-based TENG was further validated by measuring the root mean square power (P_{RMS}) across load resistance ranging from 10 Ω to 20 MΩ, as shown in Figure 3c. The P_{RMS} was calculated using Equation (5), based on the area and period (T_{cycle}) of the output signal, with the detailed calculation process depicted in Figure S5a (Supporting Information).

$$P_{RMS} = \sqrt{\frac{1}{T_{cycle}} \int (VI)^2 (t)\, dt} \tag{5}$$

while the Ni-mesh and Ni/SiO_2 exhibited P_{RMS} values of 9.1 μW and 10.6 μW, respectively, Ni/MoS_2 significantly increased to 19.2 μW due to the enhanced transferred charge density. The SiO_2@MoS_2-based TENG produced 28.8 μW at a load resistance of 14 MΩ, an improvement of more than threefold compared to that of the Ni-mesh. This enhanced output power was successfully utilized to charge a 100 μF capacitor, reaching a voltage of 2.8 V in 70 s, as shown in Figure 3d and Figure S5b (Supporting Information). A key advantage of the SiO_2@MoS_2 NPs is that they can be fully synthesized in nanoparticle form,

© 2025 Wiley-VCH GmbH

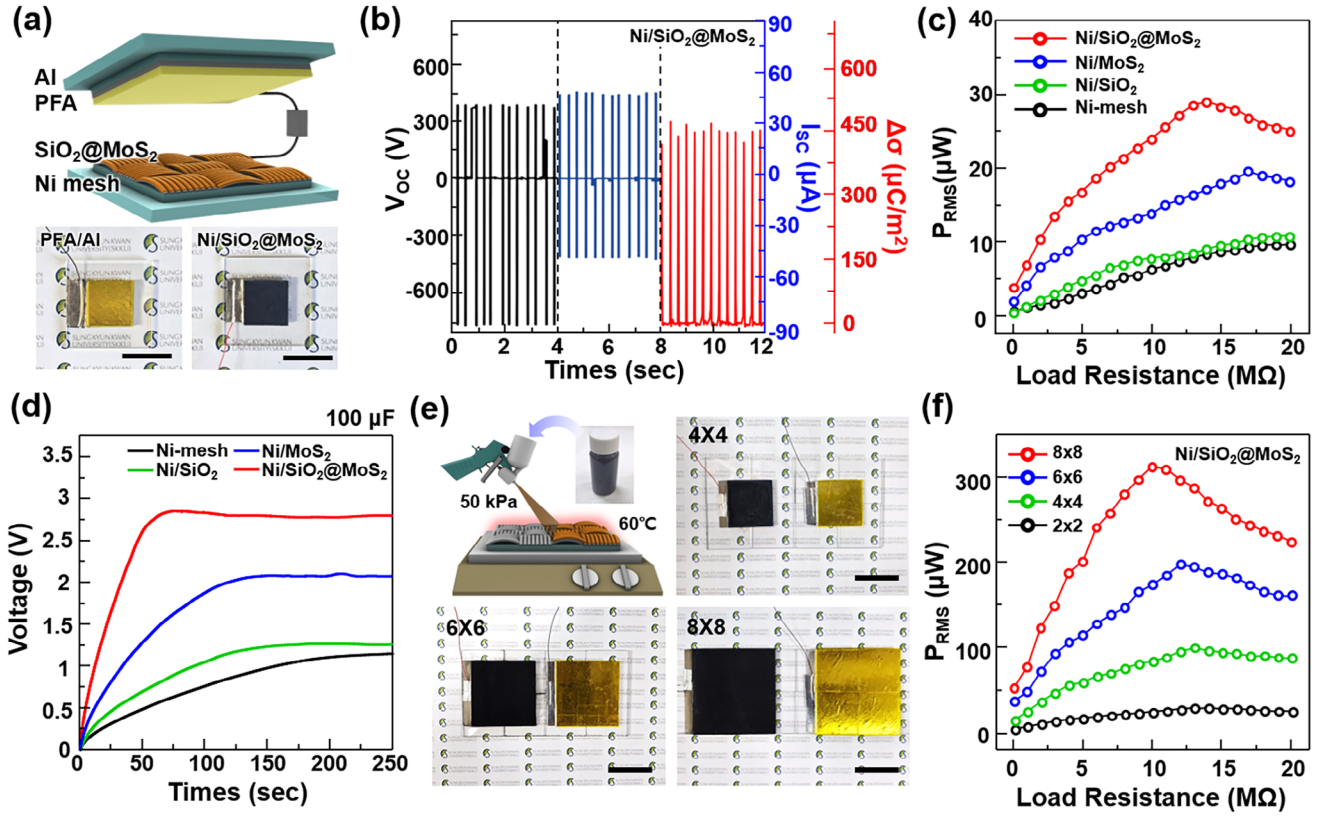


Figure 3. a) Schematic device structure and photographs of TENG, consist of PFA/Al negatively charged layer and Ni/SiO_2@MoS_2 positively charged layer. b) V_{OC}, J_{SC}, and charge density of the TENG. c) The output RMS powers of the TENGs with external loads ranging from 10 Ω to 20 MΩ. d) The charging process of 100 μF capacitor by the TENGs. e) Schematic illustration of spray-coating configuration and photographs of TENGs with increased contact area. Scale bar: 4 cm all photographs. f) The output RMS powers of the TENGs with external loads ranging from 10 Ω to 20 MΩ.

allowing their application in various manufacturing techniques such as spin-coating, spray-coating, and dip-coating, regardless of conformation of substrate. To demonstrate the feasibility of producing large-area devices where uniform films are typically challenging to form, spray-coating technology utilizing a compressor was employed. Figure 3e and Figure S6 (Supporting Information) illustrate the configuration of the equipment used for SiO_2@MoS_2 spray-coating. The spraying conditions were set to a valve pressure of 250 kPa and a spray pressure of 50 kPa, conducted at a height of 30 cm from the surface of the Ni-mesh. Through five repetitive spray-coating sessions, Ni/SiO_2@MoS_2 devices with areas of 4×4 cm^2, 6×6 cm^2, and 8×8 cm^2 were fabricated, as shown in Figure 3e, with the device surface continuously maintained at 60 °C to improve uniformity and adhesion. The elevated temperature of the substrate reduces the viscosity of the coated suspension and alleviates shrinkage stress, thereby mitigating the formation of localized micro-cracks and throughpores.[86,87] Moreover, this process facilitates uniform coating on rough surface by inducing rapid evaporation and reducing the surface tension and viscosity of the solvent. As shown in Figure S7 (Supporting Information), the SEM image after spray coating highlights the uniformity of the SiO_2@MoS_2 NPs over protrusions and depressions. Furthermore, the significantly reduced overall process time compared to sequential drying steps demonstrates its practical feasibility. The output charge increased with the device area, with an approximate twelve-fold enhancement in the 8×8 cm^2 Ni/SiO_2@MoS_2 TENG, which achieved a measured power of 318 μW at a load of 10 MΩ (Figure 3f; Figure S8, Supporting Information). However, the charge density and power density decreased to ≈75% and 69%, respectively, compared to the 2×2 cm^2 device, likely due to factors such as diminished coating uniformity and discrepancies in the contact area. Nevertheless, the 8×8 cm^2 device exhibited a high charging efficiency, charging a 100 μF capacitor to 7.6 V in only 27 s (Figure S9, Supporting Information). In addition, the influence of the mechanical parameters, including frequency and applied force, was further investigated across the ranges of 1 Hz to 4 Hz and 10 N to 40 N, respectively. As shown in Figures S10 and S11 (Supporting Information), the Ni/SiO_2@MoS_2 TENG demonstrated a pronounced increase in output with increasing frequency and applied force. While the output exhibited near saturation beyond 3 Hz, whereas the consistent enhancement with increasing mechanical force. Notably, an exceptional output charge density of 551 μC m^{-2} was achieved under a mechanical force of 40 N. However, such elevated force may pose potential challenges to the long-term stability of the device.

In general, the electrical outputs of TENGs tend to be unstable and significantly decrease under harsh conditions such as high humidity. This instability arises primarily from the complex interplay of atomic forces and molecular dynamics, which together

© 2025 Wiley-VCH GmbH

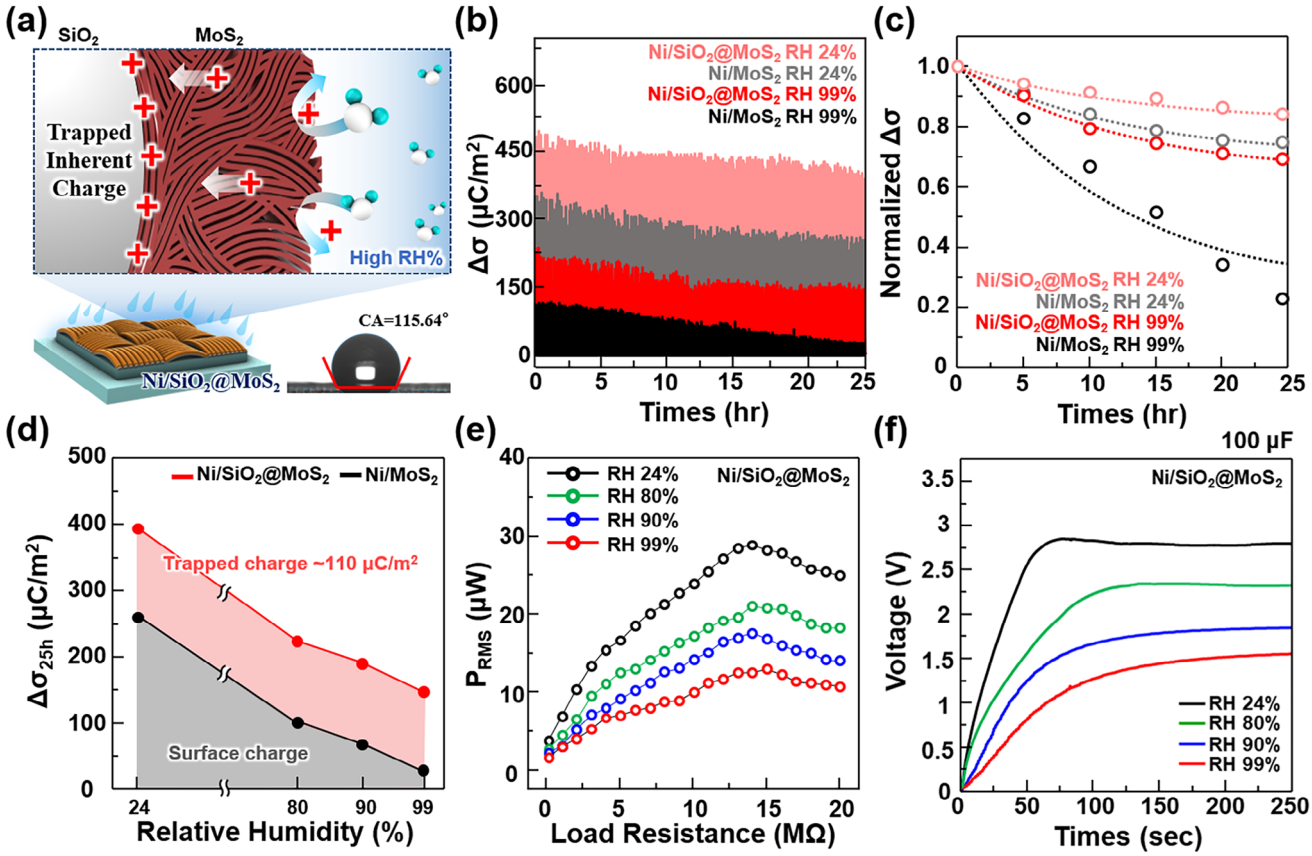


Figure 4. a) Schematic illustration of inherent trapped charges within SiO_2@MoS_2 core-shell with humidity resistance. b) The charge densities c) and the normalized charge densities of the TENGs at RH 24% and 99% as a function of operation time. d) The charge densities after 25 h operation measured at RH 24%, 80%, 90%, and 99%, with Ni/MoS_2 and Ni/SiO_2@MoS_2. e) The output RMS powers of the TENGs with external loads ranging from 10 Ω to 20 MΩ. f) The charging process of 100 μF capacitor by the TENGs.

lead to a substantial reduction in charge density.[88–90] When water molecules with dipole moments come into contact with materials where electron-nucleus binding is weakened, they can attract and capture loosely bound electrons. The partial negative charge of oxygen in these water molecules pushes the electrons away, leading to their release and a reduction in the charge density of materials. However, the inherent charge trapped within the interface is not as susceptible, resulting in low charge decay even under high RH, as illustrated in **Figure 4a**.[91] Additionally, the MoS_2 shell is predominantly hydrophobic, and the interlayer channels and marginal spaces of the flower-like MoS_2 structure provide a tortuous path for water molecule transport, as shown in Figure 2b.[92,93] This prevents water molecules from penetrating toward the core-shell interface, where inherent charges are densely trapped. The hydrophobicity of the MoS_2 shell was quantified by measuring the contact angle (CA) of a 3 μL water droplet on the surface, as shown in Figure S12 (Supporting Information). The surfaces of the Ni-mesh and Ni/SiO_2 were highly hydrophilic, whereas the CAs of Ni/MoS_2 and Ni/SiO_2@MoS_2 were ≈109.4° and 115.6°, respectively. These effects are achievable because MoS_2 is uniformly and completely formed on the surface of SiO_2.

Figure 4b demonstrates the tribo-charge density of the SiO_2@MoS_2 cationic material under different RH conditions of 24% and 99% over a 25-h period, compared to the charge density of MoS_2. The humidity was precisely regulated using a humidifier and hygrometer within a sealed acrylic box, as detailed in Figure S13 (Supporting Information). At 24% RH, the charge density of SiO_2@MoS_2 shows a relatively high initial value of ≈471 μC m^{-2}, gradually decreasing over time. In contrast, MoS_2 at the same humidity level shows a lower initial charge density but remains stable throughout the measurement. As the RH increased to 99%, the initial charge density of SiO_2@MoS_2 dropped to ≈225 μC m^{-2} (Figure 4b; Figure S14, Supporting Information). Despite this decrease, the charge density at 99% RH was still ≈2.7 times higher than that of the reference Ni-mesh at 24% RH. Notably, SiO_2@MoS_2 exhibited a remarkably low charge decay. At 99% RH, ≈67% of the initial charge density (150 μC m^{-2}) was retained after 25 h, with minimal reduction in charge density observed after 12 h. In contrast, the charge density of MoS_2 at RH 99% continuously and significantly decreased to just 31 μC m^{-2} over the same period. The superior performance and stability under high humidity conditions compared to MoS_2, was also shown in the Figure 4c. Figure 4c summarizes the normalized values of charge density measured over time at various humidity levels. The total charge density of the TENG is determined by the competition between the charge transfer rate across the contact surfaces and the charge decay rate due to triboelectric charge

© 2025 Wiley-VCH GmbH

dissipation.[94,95] To evaluate the charge decay rate, the experimental data from Figure 4c and Figure S15 (Supporting Information) were fitted to an exponential decay function as:

$$Q_{sc}\ (t) = \Delta Q_D \exp(-t/\tau) + Q_T + Q_I \quad (6)$$

where τ is a charge decay time constant. When defining the charge decay rate (ζ_1) as $\zeta_l = \Delta Q_D/Q_I$, the charge decay rates of $Ni/SiO_2@MoS_2$ were calculated to be 17.1% (RH 24%), 51.2% (RH 80%), 47.1% (RH 90%), and 33.7% (RH 99%), respectively (Figure S10, Supporting Information). Surprisingly, at higher humidity levels above RH 80%, the charge decay rate decreased as the humidity increased. This implies that interface-trapped charge is maintained with minimal sensitivity to humidity. As shown in Figure 4d and Figure S16 (Supporting Information), the difference in charge density between $Ni/SiO_2@MoS_2$ and Ni/MoS_2 was consistently ≈110 μC m^{-2} across all humidity levels and operation times, indicating that the value will be the amount of trapped inherent charges. Figure 4d suggests that the trapped charge (red-shaded area) was remained stable even after 25 h of operation at 99% RH, whereas the surface charge (gray-shaded area) decreased rapidly. Despite the significant humidity condition, the $Ni/SiO_2@MoS_2$-based TENG achieved a P_{RMS} of 12.5 μW at 15 MΩ, as shown in Figure 4e. Based on this result, the TENG successfully charged a 100 μF capacitor to 1.5 V within 190 s, even under 99% RH conditions (Figure 4f). Table 1 and Figure S17 (Supporting Information) provide a comprehensive performance comparison with prominent recent studies, highlighting charge density under humid conditions, long-term stability, and scalability as one of the most advanced values reported in recent literature.[44,45,49,52,96–111]

For practical applications, the TENG fabricated with the $Ni/SiO_2@MoS_2$ cationic framework was utilized to develop a self-powered gas sensing system capable of operating under harsh conditions with 99% RH. The gas sensing system consists of three key components: a humidity control unit (including a hygrometer, humidifier, and sealed acrylic box), a 2 × 2 cm^2 gas sensor microarray containing four types of ZnO-based gas sensing materials, and a power supply unit comprising a TENG, rectification circuit, and capacitor. The circuit diagram, schematic illustration, and experimental setup of this configuration are depicted in Figure S18 (Supporting Information). The gas sensor microarray was fabricated on a SiO_2/Si substrate using photolithography, with detailed fabrication steps provided in the Experimental Section. The four components of the microarray were spin-coated with different ZnO-based sensing materials: ZnO, ZnO/Au, ZnO/MoS_2, and ZnO/N-CDs, as illustrated in **Figure 5a**. ZnO was synthesized via the sol-gel method at a mild temperature of 90 °C by adding KOH to a zinc acetate solution. The average size of the synthesized ZnO sensing material, which contains both NPs and nanorods, was ≈80 nm. ZnO was then functionalized with Au, MoS_2, and nitrogen-doped carbon nanodots (N-CDs) to enhance gas selectivity. Enhanced electron transfer at the ZnO/N-CDs interface, CO oxidation facilitated by the oxygen spill-over effect of Au NPs, and improved SO_2 molecule adsorption by MoS_2 were expected to improve both sensitivity and selectivity for target gases.[112–115] The prepared sensing materials were further characterized using FE-SEM and EDS, revealing that the NPs, each smaller than 30 nm, were uniformly dispersed within the ZnO matrix (Figure 5a; Figure S19, Supporting Information).

The samples were then spin-coated onto the gas sensor microarray to investigate their response to four target gases: H_2, CO, H_2S, and SO_2. The sensors were placed in the humidity control unit and exposed to various partial pressures of the gases (H_2, SO_2, H_2S at 0.5 ppm, and CO at 1 ppm) entrained in air with a flow rate of 1000 sccm for 6 min. Figure 5b and Figures S20 and S21 (Supporting Information) show the resistance changes of each sensor during gas exposure, measured under both indoor humidity and RH 99% at a controlled temperature of 250 °C using a microheater attached to the bottom of the sensor array. Notably, all gas sensors exhibited clear resistance changes in response to each gas, regardless of humidity conditions. The response for all gases decreased at RH 99%, however, the response propensity corresponding to each sensor materials was clearly maintained (Figure S22, Supporting Information). Based on these results, the aggregated dataset was processed using principal component analysis (PCA) to distinguish the responses to the four different gases (Figure 5c). The PCA was performed using the singular value decomposition method in MATLAB, and the results were visualized as a 3D map to assess the degree of separation.[116,117] The features extracted from PCA allowed clustering based on the type of gas exposed. At a 95% confidence level, the chemi-resistive gas sensor array composed of four types of ZnO-based sensors, successfully identified the exposed gas species in quaternary mixtures of H_2, CO, H_2S, and SO_2. Additionally, the response time and recovery time of each gas sensor material were analyzed, with all functionalized ZnO-based gas sensor materials achieving response times below 250 s and recovery times below 300 s even under RH 99% conditions, as shown in Figures S20 and S21 (Supporting Information). The reproducibility of the gas sensors was evaluated by calculating the relative standard deviation (RSD) based on the mean and standard deviation of six repeated gas response cycles, as detailed in Figure S23 (Supporting Information). Each sensor component exhibited an RSD below 5%, regardless of humidity conditions or exposed gas, indicating competent reproducibility of the fabricated gas sensor array.

Finally, to demonstrate that the $SiO_2@MoS_2$-based TENG can supply sustainable energy to the gas sensor array under RH 99% humidity conditions, the capacitance charging of the TENG and current changes of the gas sensors during the sensing process were measured in real time (Figure 5d and Videos S1 and S2, Supporting Information). Under the conditions of 3 Hz and 30 N, the 100 μF capacitor was charged to 2.9 V and 2.2 V at RH 40% and RH 99%, respectively. When the switch connecting the TENG to the gas sensor array was activated, the voltage dropped to 1.8 V and 1.5 V. This behavior can be explained using the theoretical framework for capacitor charging:

$$Q = CV_b\left[1 - e^{-t/\tau}\right] \quad (7)$$

where V_b is applied voltage from energy source and τ is time constant. According to Equation (7), the maximum charging voltage of the capacitor $Q_{max} = CV_b$ is proportional to the total capacitance of connected devices. Therefore, when an additional gas sensor is connected in series, the total capacitance decreases, resulting in a lower maximum charging voltage. Simultaneously,

© 2025 Wiley-VCH GmbH

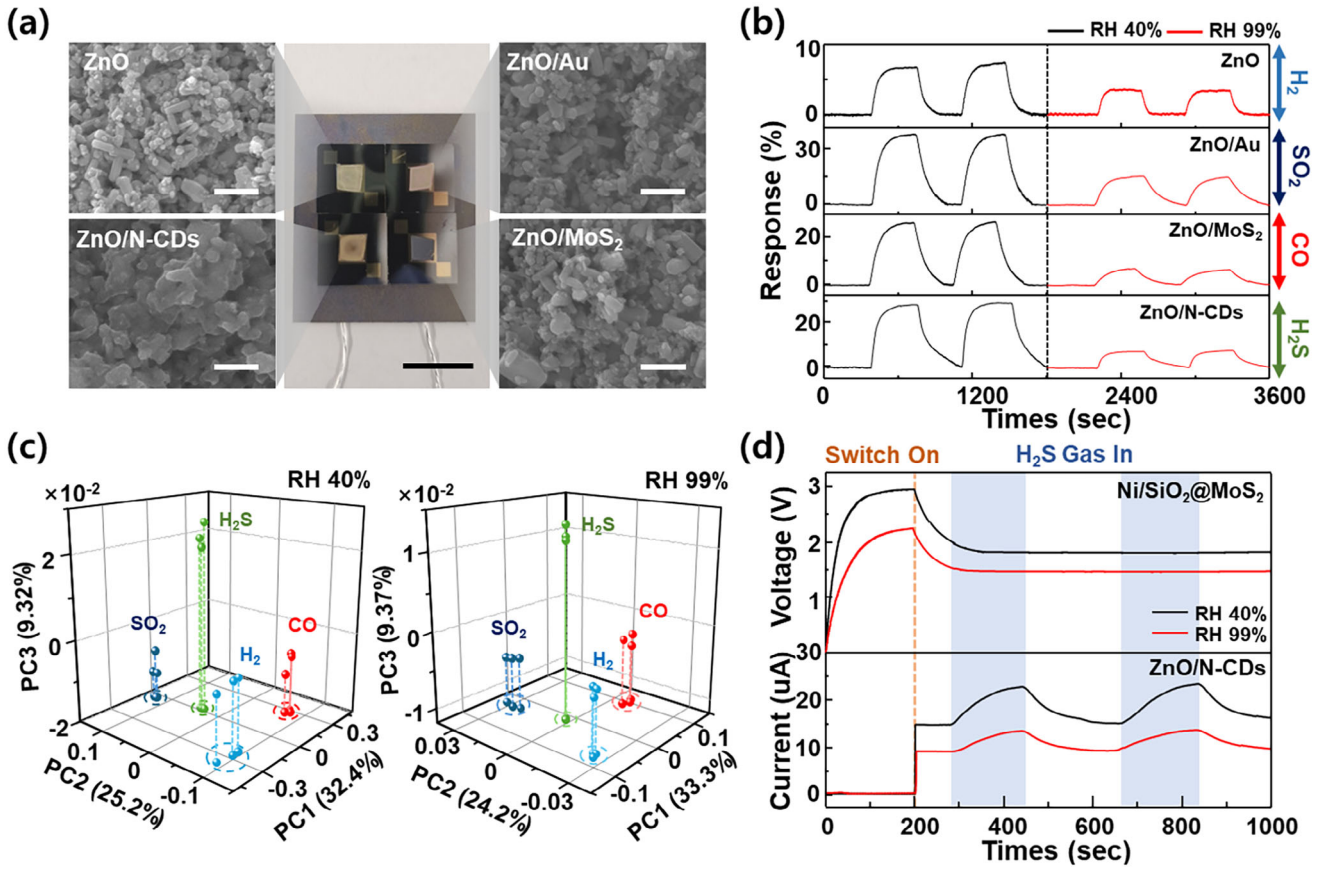


Figure 5. a) Photograph of integrated gas sensor array and SEM images of coated ZnO-based sensor materials. Scale bar: 100 nm all SEM images and 1 cm photograph. b) Gas response of ZnO, ZnO/Au, ZnO/MoS_2, and ZnO/N-CDs for H_2, SO_2, CO, and H_2S gas at 250 °C, respectively. c) Principal component analysis, plotted as 3D maps. d) The charging process of the 100 µF capacitor by the TENG before and after connecting the gas sensor array as a function of RH and the corresponding change in currents of ZnO/N-CDs gas sensor.

when the switch between the TENG and the ZnO/N-CDs gas sensor was connected, the TENG provided currents of ≈15 µA at RH 40% and 8.9 µA at RH 99%. In the absence of gas exposure, the resistance of the gas sensor remains constant, resulting in a stable current flow under a constant voltage on the capacitor. The sensor array exhibited excellent response-recovery characteristics when exposed to H_2S gas. Despite the gas exposure, the charging voltage of the capacitor remained stable at 1.8 V and 1.5 V, indicating that the TENG supplied sufficient and continuous energy to meet the power demands of the sensor system. In addition, this indicates that the capacitance of the ZnO-based sensor material remains nearly unchanged upon gas exposure, ensuring that the total capacitance of the circuit stays constant.

3. Conclusion

In conclusion, the development of SiO_2@MoS_2 core-shell NPs represents a major advancement in the performance and stability of TENGs under high humidity conditions. This innovative core-shell structure, achieved without multilayer stacking or additional packaging, utilizes a hydrophobic MoS_2 shell uniformly grown on a SiO_2 core. The growth mechanism involves selective nucleation of MoS_2 on the SiO_2 surface, where molybdenum species are reduced and sulfided to form a stable, continuous MoS_2 layer. This structure enhances charge transfer efficiency and minimizes charge decay under high humidity by isolating the SiO_2 interface from water molecules. The resulting TENG, incorporating SiO_2@MoS_2 NPs, demonstrated exceptional stability and performance, effectively powering a self-sustaining gas sensing system capable of distinguishing gases even in harsh environments with RH 99%. These findings highlight the potential of SiO_2@MoS_2-based TENGs as robust power sources for self-powered sensing applications and wearable devices, particularly in high-humidity or challenging conditions.

4. Experimental Section

Synthesis of SiO_2@MoS_2 Core-Shell Nanoparticles: All reagents for the synthesis were obtained from Sigma-Aldrich. SiO_2 nanoparticles (NPs) were prepared using the conventional Stöber method. Tetraethyl orthosilicate (TEOS, 98%) was added to a mixed solution of ethanol (EtOH, ≥99.8%), ultrapure water (18.2 MΩ), and ammonium hydroxide (NH_4OH, 28.0–30.0%). The solution was heated to 70 °C and stirred for 3 h to form SiO_2 NPs with a size of 150–200 nm.

The SiO_2@MoS_2 core-shell NPs were synthesized through a simple one-step hydrothermal process. SiO_2 NPs were dispersed in a 1:1 solution of EtOH and DI water with ammonium molybdate tetrahydrate (99.98%) and L-cysteine (98%), followed by ultrasonication for 30 min. The mixture

© 2025 Wiley-VCH GmbH

was then transferred into a 100 mL stainless steel-Teflon-lined autoclave reactor and placed in an electric oven. The temperature was increased to 180 °C at a ramp rate of 3 °C/min and maintained for 24 h. The resulting sediment was washed with EtOH five times using centrifugation (13000 rpm). Following a calcination step at 600 °C for 2 h at a ramp rate of 3 °C/min to remove organic material and moisture, SiO_2@MoS_2 core-shell NPs were obtained.

Fabrication of Triboelectric Nanogenerator (TENG): The supporting layers for both the bottom and top plates were made from acrylic plates cutted with a laser cutter (XTL US Inc., model xTool-P2). The top plate was constructed using a 2 × 2 cm^2 layer of 25 µm-thick perfluoroalkoxy alkane (PFA) film and an aluminum (Al) layer, serving as the tribo-negative material and electrode, respectively. The Al top electrode and PFA film were sequentially attached to the supporting layer. For the bottom plate, a nickel (Ni) mesh film was prepared by electroplating Ni onto polystyrene fibers. A 2 × 2 cm^2 layer of Ni mesh was first placed on the surface of the bottom supporting layer. SiO_2@MoS_2 NPs were then dispersed in ethanol and coated onto the Ni mesh by spin-coating at 1500 rpm. The resulting Ni/SiO_2@MoS_2 was obtained through annealing at 60 °C for 3 h in ambient atmosphere. During the annealing process, the Ni/SiO_2@MoS_2 was attached to the framework as the tribo-positive material.

For the fabrication of large-scale TENG devices (4 × 4 cm^2, 6 × 6 cm^2, and 8 × 8 cm^2), the SiO_2@MoS_2 solution was coated onto the Ni mesh using a spray-coating process. The spray conditions were set to a valve pressure of 250 kPa and a spray pressure of 50 kPa. The Ni mesh was placed on a hot plate set to 60 °C, and the spray was applied from a height of 30 cm. The Ni/SiO_2@MoS_2 was achieved by repeating for a total of five cycles, with a 30-min interval between each cycle.

Materials Characterization and Output Measurement of TENGs: To gain detailed insights into the morphology and structural characteristics of the synthesized NPs, field emission scanning electron microscopy (FE-SEM, JSM-IT800, JEOL Ltd.) and high-resolution transmission electron microscopy (HR-TEM, JEM-F200, JEOL Ltd.) equipped with energy-dispersive X-ray spectroscopy were employed. The band gap analysis was conducted using UPS (XPS(ESCA)/UPS, Prevac) and low-energy inverse photoemission spectroscopy (LEIPS) with a PHI 5000 VersaProbe system. For the UPS measurements, helium I (He I) was used as the excitation source, providing photon energy of 21.22 eV. The pass energy was set to 0.58 eV for work function determination and 5.58 eV for valence band (VB) measurements. A clean Ag (111) single crystal was used as the reference for Fermi level calibration. In the LEIPS measurements, a BaO cathode electron gun served as the excitation source, with a bandpass filter energy of 4.77 eV.

The open-circuit voltage (V_{OC}) and short-circuit current (J_{SC}) of TENGs were measured using a Tektronix DPO 3052 Digital Phosphor Oscilloscope and a low-noise current preamplifier (model SR570, Stanford Research Systems, Inc.), respectively, with a Labworks Inc. pushing tester (model ET-126-4) for applied force. The charge density of the TENG was evaluated using a Keithley 6514 system electrometer.

Synthesis of Gas Sensing Materials: All reagents used for the synthesis were purchased from Sigma-Aldrich. ZnO NPs were synthesized via a simple sol-gel process. Zinc acetate ($Zn(CH_3CO_2)_2$, ≥ 99.0%) and potassium hydroxide (KOH, ≥ 85%) were separately dissolved in ultrapure water (18.2 MΩ) and stirred under ambient conditions. The potassium hydroxide solution was added dropwise to the zinc acetate solution with continuous stirring until the solution became viscous and white. The mixture was then heated at 90 °C without stirring for 3 h. The resulting suspension was centrifuged to collect the product, which was then washed with deionized water and dried at 60 °C. Colloidal dispersions of Au NPs were prepared by reducing tetrachloroauric acid ($HAuCl_4$) with a cooled sodium borohydride ($NaBH_4$, 99%) solution. The solution was vigorously stirred, then washed with deionized water and ethanol, followed by drying at 60 °C overnight. MoS_2 NPs were synthesized using a heat-up method. Oleylamine and ammonium tetrathiomolybdate ($(NH_4)_2MoS_4$, 99.97%) were sequentially added to a three-necked round-bottom flask and stirred under vacuum and N_2 atmosphere at 110 °C for 30 min. The temperature was then increased to 360 °C and maintained for 30 min before cooling to room temperature. The resulting product underwent solvent-antisolvent purification using hexane and ethanol to remove byproducts. N-CDs were synthesized by a hydrothermal method. Dopamine hydrochloride (≥ 98%) and citric acid (≥ 99.5%) were dissolved in ethanol, and the solution was placed in a stainless steel-Teflon autoclave reactor. The autoclave was heated to 180 °C for 6 h and then centrifuged at 10000 rpm. The supernatant was removed, and the solution was filtered using a 0.22 µm syringe filter to obtain an ethanolic solution of N-CDs. ZnO NPs were stirred overnight with solutions of Au NPs, MoS_2, and N-CDs to produce ZnO/Au, ZnO/MoS_2, and ZnO/N-CDs heterogeneous composite gas-sensing materials, respectively. The resulting suspensions were stored for further use.

Fabrication of Gas Sensors and Response Test: To fabricate the gas sensors, a p-type SiO_2/Si wafer was prepared and spin-coated with a negative photoresist (PR, AZ2035) at 3000 rpm in a cleanroom environment. The PR-coated substrate was then pre-baked on a hotplate at 110 °C for 60 s and placed in a mask aligner (model MDA-400S, MIDAS Inc.) with a 5 × 5 square inch chrome/sodalime mask. The substrate was exposed for 8.5 s and post-baked under the same conditions as the pre-bake. After development with MIF300 developer, the substrate was cleaned with acetone and deionized water. A layer of 20 nm titanium (Ti) followed by 200 nm gold (Au) was deposited on the PR-masked substrate using an e-beam evaporator. A lift-off process was then performed by ultrasonication in acetone for ≈10 min to remove the PR coating. Finally, the patterned Au/Ti/SiO_2/Si substrate was rinsed with deionized water and dried quickly using an air gun. For deposition of the gas-sensing material, the particles were dispersed in ethanol and uniformly sonicated for 30 min. A 1 wt.% suspension of each particle was prepared, and 25 µL was dropped onto the substrate and spin-coated 20 times at 3000 rpm. The coated substrate was then heated on a hot plate at 350 °C for 30 min to remove residual solvent. The sample was subsequently annealed for 90 s using rapid thermal annealing (RTA) at 300 °C to enhance ohmic contact between the particles and metal electrodes. The response of the sensors was defined as $R(\%) = ((R_a\text{-}R_g)/R_a) \times 100(\%)$, where R_g and R_a represent the steady-state resistance values measured when the analyte gas is present and when only air is present, respectively.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by a National Research Foundation of Korea (NRF) grant, funded by the Korean government Ministry of Science and ICT (MSIT) (No. RS-2024-00448865 and RS-2023–00256847).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

humidity resistance, interfacial charge trap, MoS_2-encapsulated SiO_2 nanoparticles, self-powered sensing, triboelectric nanogenerators

© 2025 Wiley-VCH GmbH

Received: November 12, 2024
Revised: December 27, 2024
Published online: January 24, 2025

[1] S. Yun, Y. Zhang, Q. Xu, J. Liu, Y. Qin, *Nano Energy* **2019**, *60*, 600.
[2] J. Singh, R. Kaur, D. Singh, *Int. J. Energy Res.* **2021**, *45*, 118.
[3] J. He, K. Li, L. Jia, Y. Zhu, H. Zhang, J. Linghu, *Appl. Therm. Eng.* **2024**, *236*, 121813.
[4] L. Wei, H. Liu, L. Shu, L. Zhao, Z. Liu, Y. Chang, *Sens. Actuators A: Phys.* **2023**, *361*, 114577.
[5] Y. Qin, X. Wang, Z. L. Wang, *Nature* **2008**, *451*, 809.
[6] N. Sezer, M. Koç, *Nano Energy* **2021**, *80*, 105567.
[7] L.-C. Zhao, H.-X. Zou, K.-X. Wei, S.-X. Zhou, G. Meng, W.-M. Zhang, *Adv. Energy Mater.* **2023**, *13*, 2300557.
[8] H. Yu, N. Li, N. Zhao, *Adv. Energy Mater.* **2021**, *11*, 2002646.
[9] Q. Zheng, Q. Tang, Z. L. Wang, Z. Li, *Nat. Rev. Cardiol.* **2021**, *18*, 7.
[10] Q. Zhang, L. Li, T. Wang, Y. Jiang, Y. Tian, T. Jin, T. Yue, C. Lee, *Nano Energy* **2021**, *90*, 106501.
[11] T. Cheng, J. Shao, Z. L. Wang, *Nat. Rev. Methods Primers* **2023**, *3*, 39.
[12] S. Niu, Z. L. Wang, *Nano Energy* **2015**, *14*, 161.
[13] W.-G. Kim, D.-W. Kim, I.-W. T. J.-K. Kim, M.-S. Kim, Y.-K. Choi, *ACS Nano* **2021**, *15*, 258.
[14] F.-R. Fan, Z.-Q. Tian, Z. L. Wang, *Nano Energy* **2012**, *1*, 328.
[15] J. Zhao, Y. Shi, *Adv. Funct. Mater.* **2023**, *33*, 2213407.
[16] Y. Zi, S. Niu, J. Wang, Z. Wen, W. Tang, Z. L. Wang, *Nat. Commun.* **2015**, *6*, 8376.
[17] M. Chi, C. Cai, Y. Liu, S. Zhang, T. Liu, G. Du, X. Meng, B. Luo, J. Wang, Y. Shao, S. Wang, S. Nie, *Adv. Funct. Mater.* **2024**, *34*, 2411020.
[18] Y. Liang, X. Xu, L. Zhao, C. Lei, K. Dai, R. Zhuo, B. Fan, E. Cheng, M. A. Hassan, L. Gao, X. Mu, N. Hu, C. Zhang, *Small* **2024**, *20*, 2308469.
[19] B. Zhang, G. Tian, D. Xiong, T. Yang, F. Chun, S. Zhong, Z. Lin, W. Li, W. Yang, *Research* **2021**, *2021*, 7189376.
[20] C. Zhang, X. Lin, N. Zhang, Y. Lu, Z. Wu, G. Liu, S. Nie, *Nano Energy* **2019**, *66*, 104126.
[21] S. H. Chung, J. Chung, M. Song, S. Kim, D. Shin, Z. H. Lin, B. Koo, D. Kim, J. Hong, S. Lee, *Adv. Energy Mater.* **2021**, *11*, 2100936.
[22] Q. Wang, D. Hu, X. Huang, Z. Yuan, L. Zhong, Q. Sun, F. Wang, S. Xu, S. Chen, *Adv. Funct. Mater.* **2024**, *34*, 2409088.
[23] G. Cheng, Z.-H. Lin, L. Lin, Z.-l. Du, Z. L. Wang, *ACS Nano* **2013**, *7*, 7383.
[24] H. J. Hwang, D. Kwon, H.-Y. Kwon, M. Shim, J. M. Baik, D. Choi, *Adv. Energy Mater.* **2025**, *15*, 2400481.
[25] W. Kim, K. D. Pham, D. Choi, *Nano Energy* **2023**, *106*, 108088.
[26] L. Kong, Z. Fang, T. Zhang, Z. Zhang, Y. Pan, D. Hao, J. Chen, L. Qi, *Adv. Energy Mater.* **2023**, *13*, 2301254.
[27] Z. Che, S. O'Donovan, X. Xiao, X. Wan, G. Chen, X. Zhao, Y. Zhou, J. Yin, J. Chen, *Small* **2023**, *19*, 2207600.
[28] D.-S. Zhang, W.-Z. Song, L.-X. Wu, C.-L. Li, T. Chen, D.-J. Sun, M. Zhang, T.-T. Zhang, J. Zhang, S. Ramakrishna, Y.-Z. Long, *Nano Energy* **2023**, *110*, 108313.
[29] Y. Wang, J. Zhang, X. Jia, M. Chen, H. Wang, G. Ji, H. Zhou, Z. Fang, Z. Gao, *Nano Energy* **2024**, *119*, 109080.
[30] Y. Chen, P. Wang, C. Wang, J. Li, M. Tan, M. Zhao, *Mater. Today Sustain.* **2023**, *24*, 100606.
[31] J. Cheng, W. Ding, Y. Zi, Y. Lu, L. Ji, F. Liu, C. Wu, Z. L. Wang, *Nat. Commun.* **2018**, *9*, 3733.
[32] B. Dong, Q. Shi, T. He, S. Zhu, Z. Zhang, Z. Sun, Y. Ma, D. Kwong, C. Lee, *Adv. Sci.* **2020**, *7*, 1903636.
[33] S. W. Thomas III, S. J. Vella, G. K. Kaufman, G. M. Whitesides, *Angew. Chem., Int. Ed.* **2008**, *47*, 6654.
[34] J.-K. Kim, S. Jung, D.-H. Kim, J. Gao, Y. S. Lee, H. Park, H.-C. Song, J. M. Baik, *Adv. Energy Mater.* **2024**, *15*, 2304437.
[35] V. Nguyen, R. Yang, *Nano Energy* **2013**, *2*, 604.
[36] L. S. McCarty, G. M. Whitesides, *Angew. Chem., Int. Ed.* **2008**, *47*, 2188.
[37] B. X. Du, Z. Y. He, Q. Du, Y. G. Guo, *IEEE Trans. Dielectr. Electr. Insul.* **2016**, *23*, 134.
[38] J. P. Koski, S. G. Moore, R. C. Clay, K. A. O'Hearn, H. M. Aktulga, M. A. Wilson, J. A. Rackers, J. M. D. Lane, N. A. Modine, *J. Chem. Theory Comput.* **2022**, *18*, 580.
[39] J. Zhang, C. Boyer, Y. X. Zhang, *Small* **2024**, *20*, 2401846.
[40] J. E. Ritter, K. Conley, *Int. J. Adhes. Adhes.* **1992**, *12*, 245.
[41] K. Wang, Z. Qiu, J. Wang, Y. Liu, R. Chen, H. An, J. H. Park, C. H. Suk, C. Wu, J. Lin, T. W. Kim, *Nano Energy* **2022**, *93*, 106880.
[42] S. Panda, H. Jeong, S. Hajra, P. M. Rajaitha, S. Hong, H. J. Kim, *Sens. Actuators B: Chem.* **2023**, *394*, 134384.
[43] P. Lv, L. Shi, C. Fan, Y. Gao, A. Yang, X. Wang, S. Ding, M. Rong, *ACS Appl. Mater. Interfaces* **2020**, *12*, 15012.
[44] Q. Sun, F. Liang, G. Ren, L. Zhang, S. He, K. Gao, Z. Gong, Y. Zhang, X. Kang, C. Zhu, Y. Song, H. Sheng, G. Lu, H.-D. Yu, W. Huang, *Adv. Mater.* **2023**, *35*, 2210915.
[45] L. Zhou, Y. Gao, D. Liu, L. Liu, Z. Zhao, S. Li, W. Yuan, S. Cui, Z. L. Wang, J. Wang, *Adv. Energy Mater.* **2021**, *14*, 2101958.
[46] M. Zhu, Y. Deng, Y. Zheng, X. Hu, W. Xu, R. Xiong, C. Huang, *J. Hazard. Mater.* **2024**, *468*, 133770.
[47] H. Jiao, X. Lin, Y. Xiong, J. Han, Y. Liu, J. Yang, S. Wu, T. Jiang, Z. L. Wang, Q. Sun, *Nano Energy* **2024**, *120*, 109134.
[48] R. Xu, C. Xu, Y. Wang, L. Wang, H. Guo, H. Ma, X. Shen, S. Liu, S. Wu, B. Zhao, Y. Liu, Y. Long, W. Hu, H. Li, *Nano Energy* **2024**, *128*, 109931.
[49] Z. Peng, J. Song, Y. Gao, J. Liu, C. Lee, G. Chen, Z. Wang, J. Chen, M. K. H. Leung, *Nano Energy* **2021**, *85*, 106021.
[50] J. Shen, Z. Li, J. Yu, B. Ding, *Nano Energy* **2017**, *40*, 282.
[51] M. Zhou, F. Xu, L. Ma, Q. Luo, W. Ma, R. Wang, C. Lan, X. Pu, X. Qin, *Nano Energy* **2022**, *104*, 107885.
[52] J. Zhong, X. Hou, J. He, F. Xue, Y. Yang, L. Chen, J. Yu, J. Mu, W. Geng, X. Chou, *Nano Energy* **2022**, *98*, 107289.
[53] N. Cui, L. Gu, Y. Lei, J. Liu, Y. Qin, X. Ma, Y. Hao, Z. L. Wang, *ACS Nano* **2016**, *10*, 6131.
[54] Y. Katasho, Y. Liang, S. Murata, Y. Fukunaka, T. Matsuoka, S. Takahashi, *Sci. Rep.* **2015**, *5*, 13790.
[55] C. Jiang, X. Li, Y. Ying, J. Ping, *Small* **2021**, *17*, 2007805.
[56] Y. Feng, Y. Zheng, S. Ma, D. Wang, F. Zhou, W. Liu, *Nano Energy* **2016**, *19*, 48.
[57] J. He, Y. Xue, H. Liu, J. Li, Q. Liu, Y. Zhao, L. Mu, C. L. Sun, M. Qu, *ACS Appl. Mater. Interfaces* **2023**, *15*, 43963.
[58] K. N. Kim, S. Y. Kim, S. H. Choi, M. Lee, W. Song, J. Lim, S. S. Lee, S. Myung, *Adv. Electron. Mater.* **2022**, *8*, 2200819.
[59] J.-K. Kim, G. H. Han, S.-W. Kim, H. J. Kim, R. Purbia, D.-M. Lee, J. K. Kim, H. J. Hwang, H.-C. Song, D. Choi, S.-W. Kim, Z. L. Wang, J. M. Baik, *Energy Environ. Sci.* **2023**, *16*, 598.
[60] D. Liu, J. Liu, M. Yang, N. Cui, H. Wang, L. Gu, L. Wang, Y. Qin, *Nano Energy* **2021**, *88*, 106303.
[61] N. Wang, Y. Zheng, Y. Feng, F. Zhou, D. Wang, *Nano Energy* **2020**, *77*, 105088.
[62] W. Sun, N. Wang, J. Li, S. Xu, L. Song, Y. Liu, D. Wang, *Electrochim. Acta* **2021**, *391*, 138994.
[63] J. Wang, H. Wu, S. Fu, G. Li, C. Shan, W. He, C. Hu, *Nano Energy* **2022**, *104*, 107916.
[64] D. Yang, L. Zhang, N. Wang, T. Yu, W. Sun, N. Luo, Y. Feng, W. Liu, D. Wang, *Adv. Funct. Mater.* **2024**, *34*, 2306702.
[65] Q. Zhou, K. Lee, K. N. Kim, J. G. Park, J. Pan, J. Bae, J. M. Baik, T. Kim, *Nano Energy* **2019**, *57*, 903.
[66] X. Yin, Z. Chen, H. Chen, Q. Wang, Q. Chen, C. Wang, C. Ye, *EcoMat* **2024**, *6*, e12493.

 © 2025 Wiley-VCH GmbH

[67] D. W. Kim, J. H. Lee, I. You, J. K. Kim, U. Jeong, *Nano Energy* **2018**, *50*, 192.
[68] J. Li, F. Zhou, D. Min, S. Li, R. Xia, *IEEE Trans. Dielectr. Electr. Insul.* **2015**, *22*, 1723.
[69] G. Chen, Z. Xu, *J. Appl. Phys.* **2009**, *106*, 123707.
[70] K. M. Jim, K. J. Kim, Y. N. Jang, *Ind. Eng. Chem. Res.* **2013**, *52*, 11151.
[71] L. Kang, H. Fu, X. Cao, Q. Shi, J. Yao, *J. Am. Chem. Soc.* **2011**, *133*, 1895.
[72] Y. Liu, X. Li, T. Liu, Z. Zheng, Q. Liu, Y. Wang, Z. Qin, H. Guo, Y. Liang, *J. Membr. Sci.* **2022**, *659*, 120777.
[73] J. Ren, S. Song, A. Lopez-Valdivieso, J. Shen, S. Lu, *J. Colloid. Interface Sci.* **2001**, *238*, 279.
[74] W. Chen, J. Gu, Q. Liu, R. Luo, L. Yao, B. Sun, W. Zhang, H. Su, B. Chen, P. Liu, D. Zhang, *ACS Nano* **2018**, *12*, 308.
[75] S. Duraisamy, A. Ganguly, P. K. Sharma, J. Benson, J. Davis, P. Papakonstantinou, *ACS Appl. Nano Mater.* **2021**, *4*, 2642.
[76] A. D. Marinov, L. B. Priegue, A. R. Shah, T. S. Miller, C. A. Howard, G. Hinds, P. R. Shearing, P. L. Cullen, D. J. L. Brett, *ACS Nano* **2023**, *17*, 5163.
[77] C. Tan, X. Cao, X.-J. Wu, Q. He, J. Yang, X. Zhang, J. Chen, W. Zhao, S. Han, G.-H. Nam, M. Sindoro, H. Zhang, *Chem. Rev.* **2017**, *117*, 6225.
[78] Y. Guo, X. Fu, Z. Peng, *J. Nanomater.* **2017**, *6865282*, 6.
[79] S. Choi, Z. Shaolin, W. Yang, *J. Korean Phys. Soc.* **2014**, *64*, 1550.
[80] Y. Li, C. Y. Xu, L. Zhen, *Appl. Phys. Lett.* **2013**, *102*, 143110.
[81] J. Y. Park, S. Kwon, J. H. Kim, *Adv. Mater. Interfaces* **2014**, *1*, 1300089.
[82] H. Li, Y. Sun, Y. Su, R. Li, H. Jiang, Y. Xie, X. Ding, X. Wu, Y. Tang, *Nano Energy* **2021**, *89*, 106423.
[83] Z. Chen, S. Yang, J. Huang, Y. Gu, W. Huang, S. Liu, Z. Lin, Z. Zeng, Y. Hu, Z. Chen, B. Yang, X. Gui, *Nano-Micro Lett.* **2024**, *16*, 92.
[84] X. Ding, Y. Li, Y. Si, X. Yin, J. Yu, B. Ding, *Compos. Commun.* **2019**, *13*, 57.
[85] N. Yuan, J. Li, *Appl. Surf. Sci.* **2005**, *252*, 455.
[86] Z.-Y. Wei, H.-N. Cai, A. Tahir, W.-W. Zhang, X.-F. Li, Y. Zhang, Y.-P. Huang, *Y. Liu, Ceram. Int.* **2019**, *45*, 19829.
[87] M. Ahrens, R. Vaßen, D. Stöver, S. Lampenscherf, *J. Therm. Spray Technol.* **2004**, *13*, 432.
[88] G. H. Han, S. H. Lee, J. Gao, H. S. Shin, J. W. Lee, K. J. Choi, Y. Yang, H.-C. Song, Y. Kim, J. M. Baik, *Nano Energy* **2023**, *112*, 108428.
[89] V. U. Somkuwar, H. Garg, S. K. Maurya, B. Kumar, *ACS Appl. Electron. Mater.* **2024**, *6*, 931.
[90] J. Du, C. Jiao, C. Li, Y. Tao, J. Lu, Y. Cheng, X. Xia, M. Tan, H. Wang, *Nano Energy* **2023**, *116*, 108803.
[91] Y. Wang, X. Zhang, X. Jin, W. Liu, *Nano Energy* **2023**, *105*, 108021.
[92] L. J. Fang, J. H. Chen, J. M. Wang, W. W. Lin, X. G. Lin, Q. J. Lin, Y. S. He, *ACS Omega* **2021**, *6*, 2675.
[93] T.-J. Ko, J.-H. Hwang, D. Davis, M. S. Shawkat, S. S. Han, K. L. Rodriguez, K. H. Oh, W. H. Lee, Y. Jung, *Curr. Appl. Phys.* **2020**, *20*, 344.
[94] J. W. Lee, S. Jung, T. W. Lee, J. Jo, H. Y. Chae, K. Choi, J. J. Kim, J. H. Lee, C. Yang, J. M. Baik, *Adv. Energy Mater.* **2019**, *9*, 1901987.
[95] C. Zhang, L. Zhou, P. Cheng, X. Yin, D. Liu, X. Li, H. Guo, Z. L. Wang, J. Wang, *Appl. Mater. Today* **2020**, *18*, 100496.
[96] J. Wang, H. Li, D. Zhao, Q. Gao, X. Cheng, J. Wen, Z. L. Wang, T. Cheng, *Adv. Mater. Technol.* **2023**, *8*, 2300480.
[97] X. Yu, S. Fu, X. Zuo, J. Zeng, C. Shan, W. He, W. Li, C. Hu, *Adv. Funct. Mater.* **2022**, *32*, 2207498.
[98] T. Liu, R. Liang, H. He, Y. Zeng, Z. Hou, Y. Liu, J. Yuan, B. Luo, S. Zhang, C. Cai, S. Wang, D. Lu, S. Nie, *Nano Energy* **2023**, *112*, 108480.
[99] Z. Hou, T. Liu, M. Li, Y. Xu, X. Wu, S. Luo, C. Cai, Y. Mou, Y. Qiu, S. Nie, D. Lu, *Adv. Funct. Mater.* **2024**, *34*, 2400750.
[100] C. Gao, T. Liu, B. Luo, C. Cai, W. Zhang, J. Zhao, J. Yuan, P. Fatehi, C. Qin, S. Nie, *Nano Energy* **2023**, *109*, 108300.
[101] L. Liu, L. Zhou, C. Zhang, Z. Zhao, S. Li, X. Li, X. Yin, J. Wang, Z. L. Wang, *J. Mater. Chem. A* **2021**, *9*, 21357.
[102] P. Chen, J. An, S. Shu, R. Cheng, J. Nie, T. Jiang, Z. L. Wang, *Adv. Energy Mater.* **2021**, *11*, 2003066.
[103] L. He, C. Zhang, B. Zhang, Y. Gao, W. Yuan, X. Li, L. Zhou, Z. Zhao, Z. L. Wang, J. Wang, *Nano Energy* **2023**, *108*, 108244.
[104] C. Ye, D. Liu, X. Peng, Y. Jiang, R. Cheng, C. Ning, F. Sheng, Y. Zhang, K. Dong, Z. L. Wang, *ACS Nano* **2021**, *15*, 18172.
[105] F. Peng, M. Gao, Z. Han, D. Liu, K. Dai, L. Mi, D. Zhang, G. Zheng, C. Liu, C. Shen, *Nano Energy* **2022**, *103*, 107771.
[106] S. M. S. Rana, M. T. Rahman, M. A. Zahed, S. H. Lee, Y. D. Shin, S. Seonu, D. Kim, M. Salauddin, T. Bhatta, K. Sharstha, J. Y. Park, *Nano Energy* **2022**, *104*, 107931.
[107] N. Wang, Y. Feng, Y. Zheng, L. Zhang, M. Feng, X. Li, F. Zhou, D. Wang, *Adv. Funct. Mater.* **2021**, *31*, 2009172.
[108] Z. Li, B. Xu, J. Han, J. Huang, K. Y. Chung, *Adv. Energy Mater.* **2021**, *11*, 2101294.
[109] W. Zhang, X. Chen, J. Zhao, X. Wang, X. Li, T. Liu, B. Luo, Y. Qin, S. Zhang, M. Chi, S. Wang, S. Nie, *Nano Energy* **2023**, *108*, 108196.
[110] P. Zhang, H. Huang, X. Wang, K. Cai, J. Chen, Y. Xu, F. Yu, S. Nie, S. Wang, X. Liu, *Nano Energy* **2024**, *124*, 109472.
[111] J. Wang, Z. Xia, H. Yao, Q. Zhang, H. Yang, *ACS Appl. Mater. Interfaces* **2023**, *15*, 47208.
[112] D. Degler, S. Rank, S. Müller, H. W. Pereira de Carvalho, J.-D. Grunwaldt, U. Weimar, N. Barsan, *ACS Sens.* **2016**, *1*, 1322.
[113] R. Purbia, Y. M. Kwon, H.-D. Kim, Y. S. Lee, H. Shin, J. M. Baik, *J. Mater. Chem. A* **2020**, *8*, 11734.
[114] T. Balakrishnan, S. Sagadevan, M.-V. Le, T. Soga, W.-C. Oh, *Nanomaterials* **2024**, *14*, 11.
[115] Y. M. Kwon, B. Oh, R. Purbia, H. Y. Chae, G. H. Han, S.-W. Kim, K.-J. Choi, Y. Lee, J. J. Kim, J. M. Baik, *Sens. Actuators B: Chem.* **2023**, *375*, 132939.
[116] P.-G. Su, M.-C. Li, *Sens. Actuators A: Phys.* **2021**, *331*, 112980.
[117] A. S. R. Murthy, D. Pathak, G. Sharma, K. I. Gnanasekar, V. Jayaraman, A. M. Umarji, T. Gnanasekaran, *Anal. Chim. Acta* **2015**, *892*, 175.

 © 2025 Wiley-VCH GmbH